Henry Russom

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EDUCATION

Indiana University Bloomington, Luddy School of Informatics, Computing, and Engineering Expected May 2028
B.S. Computer Science (Artificial Intelligence specialization) and B.S. Informatics | Minors: Business, Spanish
Cumulative GPA: 3.52 / 4.00 | Informatics GPA: 3.91 / 4.00 | Dean's List (Fall 2025)
Coursework: Software Systems (Java, OOP), Discrete Structures, AI Programming, Information Infrastructure, HCI

EXPERIENCE

DMC Insurance | Fishers, IN May 2026 – Present
AI Software Development Intern (Full-Time May – Aug 2026; Part-Time Remote Aug 2026 – Present)

- Designed, built, shipped, and own end to end **DemandView**, a production **Claude API** pipeline that splits **500 to 1,300 page** scanned bodily-injury demand packets into page images and processes them in batches, carrying claim context between calls via JSON handoffs
- Replaced **4+ hours** of manual reading per demand packet with a hands-off run; in production since August 2026 with **4 adjusters** across **12 claims**, including a **1,300-page** packet
- Spent the first 2 weeks interviewing adjusters (the end users) to find workflow pain points, scoped the feature with my manager, and got the design **approved by the tech lead** before building
- Shipped a **medical treatment timeline** with severity color-coding, treatment-gap detection, and a cumulative billed-cost chart, giving adjusters a one-screen view of a claim's history
- Countered inconsistent LLM extraction by storing every page as an image and adding a **one-click source link** beside each extracted fact, so adjusters can verify any output against the original page
- Own production reliability: set up **failure alerting**, troubleshoot production issues from server logs **near-daily**, and ship fixes while maintaining technical documentation for the codebase
- Secured sensitive claim data with **row-level access controls** so each adjuster sees only their own uploads, plus a **scheduled job** that purges claim PII one year after a claim's last use
- Validate every AI-generated (Claude Code) change by reading the full diff and running it against synthetic demand-packet **test fixtures**, then confirm output with adjusters on real claims
- Built a **SQL search tool** with a guided query builder over SQL Server plus a **natural-language AI search** mode, letting non-technical staff query claim data without writing SQL
- Built an automated **claim status report generator**; write **PHP** services and design **SQL Server** schemas on **Azure**

Host with Rally | Bloomington, IN May 2026 – Aug 2026
Founding Backend Engineer

- Built the **Supabase**-backed booking pipeline in **TypeScript** for an event and ticketing platform: natural-language event intake parsing, vendor scoring, and automated vendor outreach with n8n
- Diagnosed production failures on **Cloudflare Workers** and Supabase, including **4 broken cron endpoints**, **14-migration** schema drift, and a silent commission-rate fallback, then drove the fixes
- Ran a **row-level security audit** across **76+ tables** that led the team to redesign the database

Alt Construction | Zionsville, IN Summers 2021 – 2025
Seasonal Associate

- Interpreted between English-speaking site managers and Spanish-speaking crews on residential builds

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, PHP, JavaScript
Backend & Data: Supabase (Postgres, Edge Functions, RLS), SQL Server, Cloudflare Workers, Azure (VMs, Data Lake), Pandas
AI: Claude API (vision, batch processing), OpenAI API, Azure AI Content Understanding, Claude Code, n8n
Tools: Git/GitHub, Figma

LEADERSHIP

Sigma Alpha Epsilon, Indiana University | Member 2024 – Present